WORKSHEET

Chapter 2: More complex equations

Solve each equation:

1 $4x - 1 = x + 2$

2 $x + 7 = 3x + 5$

3 $8x + 2 = 5x + 11$

4 $5a + 8 = 11x - 4$

5 $3k + 6 = 8k - 5$

6 $5x = 3(x + 4)$

7 $5(y - 6) = 3y$

8 $3(a + 2) = 5a$

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9 $12m = 5(m + 7)$

10 $8p = 3(p - 10)$

11 $3(4x + 1) = 6x + 9$

12 $2(3x + 1) = 5x + 4$

13 $7x + 6 = 5(2x + 3)$

14 $-4(3x - 2) - 4x = 8$

15 $5 + 8x = 3(7 - 2x)$

16 $3(2x + 1) = 5(x + 3)$

17 $3(x + 5) = 2(4x + 3)$

18 $6(2x - 1) = 4(5x + 2)$

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19 $3(6x + 5) = -(3x - 10)$

20 $-7(2x - 3) = 2(4x - 5)$

21 $\frac{2x}{5} + 1 = 3$

22 $\frac{4m}{7} - 2 = 2$

23 $\frac{5k}{6} + 3 = 8$

24 $\frac{2a}{7} - 5 = -9$

25 $6 - \frac{3b}{4} = 0$

26 $\frac{4x - 2}{5} = 2$

27 $\frac{3x + 2}{11} = 1$

28 $\frac{6x + 2}{4} = 5$

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29 $\frac{5x-3}{7} = 2$

30 $\frac{8-2x}{5} = 3$

31 $\frac{x+7}{3}+4=6$

32 $\frac{z-5}{2}+3=4$

33 $\frac{a+8}{3}=5+10$

34 $\frac{7-b}{6}+3=-1$

35 $\frac{10-f}{7}-6=-8$

36 $\frac{8x}{2}+3=x-6$

37 $7-\frac{x}{2} = 5x-4$

38 $\frac{2x}{3} + 1 = 4x+5$

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39 $\frac{x}{5} - 1 = 3x + 4$

40 $\frac{5x}{2} + 8 = 4x - 7$

41 $\frac{3x + 2}{5} = x + 4$

42 $\frac{2x + 1}{6} = \frac{x + 3}{2}$

43 $\frac{4x - 1}{5} = \frac{2x + 3}{6}$

44 $\frac{5x + 3}{3} = \frac{4x - 5}{2}$

45 $\frac{6 - 3x}{2} = \frac{8x + 1}{3}$

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Answers

1 1
2 1
3 3
4 2
5 $2\frac{1}{5}$
6 6
7 15
8 3
9 5
10 -6
11 1
12 2
13 -3
14 0
15 $1\frac{1}{7}$
16 12
17 $1\frac{4}{5}$
18 $-1\frac{3}{4}$
19 $\frac{-5}{21}$
20 $1\frac{9}{22}$
21 5
22 7
23 6
24 -14
25 8
26 3
27 3
28 3
29 $3\frac{2}{5}$
30 $-3\frac{1}{2}$
31 -1
32 7
33 37
34 31
35 24
36 -3
37 2
38 $-1\frac{1}{5}$
39 $-1\frac{11}{14}$
40 10
41 -9
42 -8
43 $1\frac{1}{2}$
44 $10\frac{1}{2}$
45 $\frac{16}{25}$

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